

A gauge-independent spectral gap from the two-torsion geometry of a torus quotient

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Abstract

Let (M, g) be a compact connected Riemannian manifold containing an embedded copy of the pillowcase, the quotient of the flat two-torus by the sign involution, and let Δ be the minimum geodesic distance from a distinguished corner of the pillowcase fibre to the remaining three. We prove that the corners are precisely the four two-torsion points of the torus, pairwise distinct, and that $\Delta > 0$. We further prove that the framed moduli spaces of torsion-free sheaves on the projective plane admit cellular decompositions with vanishing odd Betti numbers; that the central Fourier coefficients of the level-two Appell–Lerch sum equal the odd divisor function $d_{\text{odd}}(n)$; and that the quaternion group decomposes into four degree-two orbits under its inner automorphisms and centre, with a transit parity $(-1)^n$ characterising the level-two multiplier-weighted Kloosterman-type sum at modulus two. For a class of Kaluza–Klein-reduced theories defined by three explicit spectral hypotheses, Δ is the mass of the lowest non-vacuum one-particle state, and neither this identification nor the positivity of Δ refers to the gauge group; the independence is machine-checked, so existence transfers across gauge groups while the value depends on the corner-selection data. An existence-only result of this kind may be useful in making progress towards the mass-gap problem for four-dimensional gauge theories.

1 Introduction

The purpose of this paper is to establish, from elementary geometric and arithmetic inputs, the existence of a strictly positive spectral gap for a class of Kaluza–Klein-reduced theories, and to show that the existence claim is independent of the gauge group of the reduced theory. The results divide into an unconditional part and a conditional part, and the division is maintained explicitly throughout.

The unconditional part comprises five statements. First, the fixed-point set of the involution $\sigma(x, y) = (-x, -y)$ on the flat two-torus $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ consists of exactly the four two-torsion points, and these are pairwise distinct; the quotient $\mathbb{T}^2/\sigma$ is the pillowcase, and the four points are its corners (Lemma 2.2). Second, if the pillowcase is embedded as a fibre of a compact connected Riemannian manifold (M, g) with a distinguished corner c , the minimum geodesic distance Δ from c to the remaining three corners is strictly positive (Lemma 2.3). Third, the framed moduli spaces $M(r, n)$ of torsion-free sheaves on the projective plane admit a Białynicki-Birula decomposition into affine cells, so that all odd Betti numbers vanish and the Euler characteristic equals the absolute Betti count (Theorem 5.1). Fourth, the central Fourier coefficients of the level-two Appell–Lerch sum equal the odd divisor function $d_{\text{odd}}(n)$ (Proposition 3.4); through the generating identity of Göttsche and the resolution of instanton moduli by Hilbert schemes of points, coefficients of this kind organise the counting invariants of the moduli spaces of the third statement. Fifth, the quaternion group Q_8 decomposes into exactly four degree-two orbits under the action generated by its inner automorphisms and

its centre, every pure imaginary unit quaternion squares to -1 , and the level-two multiplier-weighted Kloosterman-type sum at modulus two equals $(-1)^n$ for all $n \geq 1$ if and only if its single multiplier value equals -1 (Lemma 4.2, Proposition 4.6).

The conditional part concerns a class $\mathcal{C}$ of Kaluza–Klein-reduced theories, defined in Section 6 by three explicit spectral hypotheses: the level degeneracies of a theory in the class equal $d_{\text{odd}}(n)$; the mass at level n equals $n\Delta$; and $d_{\text{odd}}(n)$ counts one-particle states at level n . The hypotheses are stated as part of the definition of the class and are not derived here; the unconditional results ground the counting object to which they refer. For every $\mathcal{T} \in \mathcal{C}$ we prove that Δ is the mass of the lowest non-vacuum one-particle state (Proposition 7.1), and that neither this identification nor the positivity of Δ makes reference to the gauge group of $\mathcal{T}$ (Proposition 7.2). The latter statement is verified mechanically: in the accompanying formal development the gauge group enters as a parameter that provably does not occur in the proof term. A corollary delimits the scope: the existence of the gap is uniform over gauge groups and over admissible corner selections, whereas the numerical value of Δ is a function of the selection data and is not a gauge-group invariant (Corollary 8.1). A parallel conditional statement is geometric rather than spectral: under two declared identifications — that each half-period translation of the torus lifts to the fibre algebra as right-multiplication by a pure imaginary unit quaternion, and that one full covering period of the torus comprises two half-period translations — the cumulative fibre phase over n covering periods is $(-1)^n$ (Proposition 4.3); the identifications are declared, not derived, in exact parallel with the class hypotheses.

The method is deliberately elementary. The positivity of Δ consumes only the two-torsion structure of σ and the Hopf–Rinow theorem. The homological statement rests on the theorem of Białynicki-Birula together with the torus-action analysis of the framed moduli spaces due to Nakajima and Yoshioka. The arithmetic statement is a computation with geometric series occupying half a page, and the quaternionic statements are finite checks on an eight-element group together with a single-term evaluation. The conditional results are exact inferences from the stated hypotheses, with the single internal input $\Delta \geq 0$ discharged by Lemma 2.3. All results in the certified chain, together with the homological parity implication, the Appell–Lerch coefficient identity, the orbit decomposition and transit parity, and the multiplier characterisation, are accompanied by machine-checked certificates, reproduced in Appendix A.

The paper is organised as follows. Section 2 states the geometric setting and proves the two lemmas on the corner gap. Section 3 defines the odd divisor function, establishes its elementary properties, and proves the Appell–Lerch coefficient identity with a table to order fifteen. Section 4 proves the orbit decomposition of the quaternion group, the transit parity under the two declared identifications, and the characterisation of the multiplier-weighted sum at modulus two. Section 5 proves the vanishing of the odd Betti numbers of the framed moduli spaces and assembles the generating-function link between the moduli invariants and the coefficients of Section 3. Section 6 defines the theory class and records a remark on a second, modularity-conditional route to the absence of a paired sector. Section 7 proves the identification of Δ with the lowest mass and the gauge-group independence of the argument. Section 8 delimits the scope of the results and closes with an outlook. Appendix A reproduces the formal certificates.

2 The geometric setting and the corner gap

This section fixes the geometric data and proves the two unconditional lemmas on which the positivity of the gap rests. Both proofs are elementary; we work the smallest cases by hand.

Definition 2.1 (Setting). Let $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ be the flat two-torus and let $\sigma: \mathbb{T}^2 \rightarrow \mathbb{T}^2$ be the involution $\sigma(x, y) = (-x, -y)$. The quotient $\mathbb{T}^2/\sigma$ is the *pillowcase*. Let (M, g) be a compact connected Riemannian manifold with continuous metric, and let $\iota: \mathbb{T}^2/\sigma \rightarrow M$ be a continuous injection of the pillowcase onto a fibre of M . Let $c \in M$ be the image under ι of one distinguished corner of the pillowcase (the *selection*), and define

$$\Delta = \Delta(g, \iota, c) = \min_{c'} \text{dist}_M(c, c'), \quad (1)$$

the minimum being taken over the images c' of the three remaining corners.

The hypotheses of Definition 2.1 are the entire geometric input of the paper. Which corner is distinguished is data, not a consequence of the topology; the dependence of the value of Δ on this datum is the subject of Corollary 8.1.

Lemma 2.2 (Corner distinctness). *The fixed-point set of σ consists of exactly the four two-torsion points*

$$\text{Fix}(\sigma) = \left\{ (0, 0), \left(\frac{1}{2}, 0\right), \left(0, \frac{1}{2}\right), \left(\frac{1}{2}, \frac{1}{2}\right) \right\},$$

and these four points are pairwise distinct in $\mathbb{T}^2$. Consequently the pillowcase has exactly four corners, and their images under ι are four distinct points of M .

Proof. We work with representatives $(x, y) \in [0, 1)^2$, the fundamental domain of $\mathbb{R}^2/\mathbb{Z}^2$.

Fixed points are two-torsion. A point $[(x, y)]$ is fixed by σ if and only if $(-x, -y) \equiv (x, y) \pmod{\mathbb{Z}^2}$, that is, if and only if $2x \in \mathbb{Z}$ and $2y \in \mathbb{Z}$. The two conditions are independent, so it suffices to solve $2x \in \mathbb{Z}$ for $x \in [0, 1)$.

The smallest case, worked by hand. If $2x = k \in \mathbb{Z}$ with $x \in [0, 1)$, then $k = 2x \in [0, 2)$, so $k \in \{0, 1\}$ and correspondingly $x \in \{0, \frac{1}{2}\}$. Both solve the congruence: $-0 \equiv 0$ and $-\frac{1}{2} \equiv \frac{1}{2} \pmod{\mathbb{Z}}$, their difference being $-1 \in \mathbb{Z}$. Hence the fixed-point set is exactly $\{0, \frac{1}{2}\} \times \{0, \frac{1}{2}\}$, the four points displayed.

Distinctness. Two representatives in the fundamental domain $[0, 1)^2$ define the same point of $\mathbb{T}^2$ if and only if they are equal as pairs; the four listed representatives are pairwise unequal, their coordinatewise differences being $\pm\frac{1}{2} \notin \mathbb{Z}$ in at least one coordinate. Hence the four fixed points are pairwise distinct in $\mathbb{T}^2$. The corners of the pillowcase are by definition the images of the fixed points in $\mathbb{T}^2/\sigma$, on which the quotient map is injective; and ι is injective by hypothesis, so the four corner images are distinct in M . $\square$

We note that Lemma 2.2 is a statement of torus topology: no structure on the fibre beyond the involution is used, and no property of any group action other than σ enters.

Lemma 2.3 (Positivity of the corner gap). *In the setting of Definition 2.1, each of the three distances $\text{dist}_M(c, c')$ is strictly positive, and hence $\Delta > 0$.*

Proof. By Lemma 2.2 the four corner images are pairwise distinct points of M . By the Hopf–Rinow theorem [5, Ch. 7], a compact Riemannian manifold with continuous metric is a complete metric space in the geodesic distance, and the distance between two distinct points is strictly positive. Each of the three distances in (1) is therefore positive, and the minimum of three positive numbers is positive. Neither step invokes any group beyond the involution σ . $\square$

The formal certificate of Lemma 2.3 (Appendix A, Certificate 2) is proved at metric-space level, which is strictly sufficient for the claim as used; the Riemannian-geometric source of the metric-space structure is the Hopf–Rinow theorem as cited.

3 The odd divisor function and the Appell–Lerch sum

This section supplies the arithmetic ingredient. We define the odd divisor function, record the elementary properties consumed later, and prove that it arises as the sequence of central Fourier coefficients of the level-two Appell–Lerch sum.

Definition 3.1. For $n \geq 1$, let $d_{\text{odd}}(n)$ denote the number of odd divisors of n . Writing $n = 2^a n_{\text{odd}}$ with n_{odd} odd, one has $d_{\text{odd}}(n) = d(n_{\text{odd}})$, the ordinary divisor count of the odd part.

Proposition 3.2 (Elementary properties). *The function d_{odd} satisfies:*

- (i) $d_{\text{odd}}(1) = 1$;
- (ii) $d_{\text{odd}}(n) \geq 1$ for all $n \geq 1$;
- (iii) $d_{\text{odd}}(2^a m) = d_{\text{odd}}(m)$ for all $a \geq 0$ and $m \geq 1$;
- (iv) $d_{\text{odd}}(mn) = d_{\text{odd}}(m) d_{\text{odd}}(n)$ whenever $\gcd(m, n) = 1$;
- (v) $d_{\text{odd}}(n) = 1$ if and only if n is a power of two, and $d_{\text{odd}}(n) = 2$ if and only if n_{odd} is prime.

Proof. The sole divisor of 1 is 1, which is odd, giving (i); the divisor 1 of any n is odd, giving (ii); and (iii) is immediate from $d_{\text{odd}}(n) = d(n_{\text{odd}})$, since $2^a m$ and m have the same odd part for odd m , and in general the odd part is unchanged by powers of two. Property (iv) follows from the multiplicativity of d together with the multiplicativity of the odd-part map on coprime arguments. For (v), $d(n_{\text{odd}}) = 1$ if and only if $n_{\text{odd}} = 1$, and $d(n_{\text{odd}}) = 2$ if and only if n_{odd} is prime. $\square$

Definition 3.3. The level-two Appell–Lerch sum is

$$\mu(\tau, z) = \sum_{n \geq 1} \frac{q^n}{(1 - yq^n)(1 - y^{-1}q^n)}, \quad q = e^{2\pi i \tau}, \quad y = e^{2\pi i z}, \quad (2)$$

regarded as a formal power series in q with coefficients in Laurent polynomials in y .

Appell–Lerch sums are the canonical building blocks in Zwegers’ theory of mock theta functions [14]; the specialisation (2) is the level-two member of the family, corresponding to the order-two quotient in Definition 2.1. The property we require of μ is purely combinatorial.

Proposition 3.4 (Central coefficients). *The coefficient of $y^0 q^m$ in $\mu(\tau, z)$ equals $d_{\text{odd}}(m)$ for every $m \geq 1$.*

Proof. Expanding each factor of (2) in geometric series,

$$\frac{q^n}{(1 - yq^n)(1 - y^{-1}q^n)} = \sum_{a, b \geq 0} y^{a-b} q^{n(1+a+b)}.$$

The y^0 terms are those with $a = b$, so the central part of μ is

$$\sum_{n \geq 1} \sum_{a \geq 0} q^{n(2a+1)}.$$

The coefficient of q^m therefore counts the pairs (n, a) with $n \geq 1$, $a \geq 0$ and $n(2a + 1) = m$. The map $(n, a) \mapsto 2a + 1$ is a bijection from this set of pairs onto the set of odd divisors of m , with inverse $d \mapsto (m/d, (d - 1)/2)$. Hence the coefficient is $d_{\text{odd}}(m)$. $\square$

The bijection in the proof of Proposition 3.4 is machine-checked in full generality, and the resulting coefficient table certified to order fifteen, in Appendix A (Certificate 7). The certified table is

m	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$d_{\text{odd}}(m)$	1	1	2	1	2	2	2	1	3	2	2	2	2	2	4

in agreement with Definition 3.1 evaluated directly.

4 The quaternionic fibre and the corner-transit parity

This section supplements the arithmetic of Section 3 with a group-theoretic structure carried by the quaternion group, and establishes three results: an orbit decomposition, a transit parity, and a characterisation of the level-two multiplier-weighted Kloosterman-type sum at modulus two. The first and third are unconditional; the second is an exact inference from two declared geometric identifications, stated as explicit hypotheses in the proposition. Nothing in this section derives any hypothesis of the class of Section 6, and the relation of the present results to that class is stated there.

Definition 4.1 (The fibre algebra and its two four-groups). Let $\mathbb{H}$ denote the quaternion algebra over $\mathbb{R}$ with basis $1, i, j, k$ and relations $i^2 = j^2 = k^2 = ijk = -1$ [4], and let

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\} \subset \mathbb{H}^\times$$

be the quaternion group, with centre $Z = \{\pm 1\} = \langle -1 \rangle$. Two Klein four-groups act in what follows. The first is geometric: the two-torsion subgroup $\mathbb{T}^2[2] \subset \mathbb{T}^2$, whose underlying set is the fixed-point set of Lemma 2.2, acting on the torus by the half-period translations. The second is algebraic: the inner automorphism group $V = \text{Inn}(Q_8) \cong Q_8/Z \cong \mathbb{Z}/2 \times \mathbb{Z}/2$, generated by conjugation by i and conjugation by j , acting on the set Q_8 .

The two four-groups are isomorphic as abstract groups but act on different objects, the first on the base torus and the second on the fibre algebra; no identification between them is made anywhere in this paper, and the two actions enter the argument in logically separate roles, the translations through the covering geometry of the torus and the automorphisms through the orbit structure of Q_8 .

Lemma 4.2 (Orbit decomposition). *The action of $V \times Z$ on Q_8 — inner automorphisms together with multiplication by the centre — has exactly four orbits,*

$$\{\pm 1\}, \quad \{\pm i\}, \quad \{\pm j\}, \quad \{\pm k\},$$

each of cardinality two. The centre factor is essential: the inner automorphisms fix $+1$ and -1 separately, as does quaternionic conjugation $q \mapsto \bar{q}$, so no group of automorphisms or anti-automorphisms alone connects the orbit $\{\pm 1\}$.

Proof. Conjugation by i fixes ± 1 and $\pm i$ and sends $j \mapsto -j$ and $k \mapsto -k$; conjugation by j fixes ± 1 and $\pm j$ and sends $i \mapsto -i$ and $k \mapsto -k$. Each of the four displayed pairs is therefore closed under both generating automorphisms, and multiplication by -1 exchanges the two elements of each pair, so each pair is a single orbit of $V \times Z$; the pairs are disjoint and cover Q_8 . The smallest case, worked by hand: the orbit of $+1$. Every inner automorphism fixes the centre pointwise, and conjugation fixes $+1$ and -1 separately, so neither moves $+1$; the sole generator connecting $+1$ to -1 is the centre itself, and the orbit is $\{\pm 1\}$, of cardinality two. The full decomposition — closure under the generators, connectedness within each pair, disjointness, covering, and the four cardinalities — is machine-checked by exhaustive decision on an explicit eight-element model in Appendix A (Certificate 8). $\square$

Proposition 4.3 (Transit parity). *Every pure imaginary unit quaternion squares to -1 :*

$$(\pm i)^2 = (\pm j)^2 = (\pm k)^2 = -1.$$

Consequently, suppose the following two identifications are declared: (i) each half-period translation of $\mathbb{T}^2$ lifts to the fibre algebra as right-multiplication by a pure imaginary unit quaternion; (ii) one full covering period of $\mathbb{T}^2$ comprises two half-period translations. Then the cumulative fibre phase over n full covering periods equals $(-1)^n$ for every $n \geq 0$, independently of the choice of unit in (i).

Proof. The squares are immediate from the defining relations: $i^2 = -1$, and $(-i)^2 = (-1)^2 i^2 = -1$; likewise for j and k . For the parity, let u be the unit of identification (i). By (ii) one covering period multiplies the cumulative phase by $u^2 = -1$, so over n periods the phase is $(-1)^n$; the value does not depend on which of the six units is chosen, since all six square to -1 . The smallest case, worked by hand: $n = 1$ gives one covering period, hence two half-period lifts, hence phase $u \cdot u = -1 = (-1)^1$. The implication is machine-checked with the identifications (i) and (ii) encoded as named hypothesis fields (Appendix A, Certificate 8), in the same pattern as Certificate 3: what is proved is exactly the inference from the declared identifications, whose truth for any given reduction is a question about that reduction and is outside the scope of this paper. $\square$

We turn to the arithmetic object that carries the same parity.

Definition 4.4. Let ν be a multiplier system of weight one-half on $\Gamma_0(2)$, in the Rademacher–Zuckerman normalisation [11]. For integers $c \geq 1$ and each h with $0 \leq h < c$ and $\gcd(h, c) = 1$, let $\gamma_{h,c}$ denote the group element of that normalisation with lower-left entry c , and let $\bar{h}$ satisfy $h\bar{h} \equiv -1 \pmod{c}$. The multiplier-weighted Kloosterman-type sum is

$$K_2(n, m; c) = \sum_{\substack{0 \leq h < c \\ \gcd(h, c) = 1}} \bar{\nu}(\gamma_{h,c}) e^{2\pi i(n\bar{h} + mh)/c}. \quad (3)$$

The sums (3) are not the classical Kloosterman sums $S(n, m; c) = \sum_{x \in (\mathbb{Z}/c\mathbb{Z})^\times} e^{2\pi i(nx + m\bar{x})/c}$ [7], which are real-valued for all arguments, the terms at x and $-x$ being complex conjugates; the multiplier weights $\bar{\nu}(\gamma_{h,c})$ produce genuinely complex values in general, and the distinction is quantitative at the modulus of interest, as Proposition 4.6 records. Sums of the form (3) are the arithmetic companions of the Fourier coefficients of series of Appell type in exact expansions of Rademacher type [11, 3].

Lemma 4.5. *At $c = 2$ the sum (3) with $m = -1$ has the single term $h = 1$, with $\bar{h} = 1$, and evaluates to*

$$K_2(n, -1; 2) = \bar{\nu}(\gamma_{1,2}) e^{\pi i(n-1)} = \bar{\nu}(\gamma_{1,2}) (-1)^{n-1}. \quad (4)$$

Proof. The units modulo 2 are $\{1\}$, so the sum has the single term $h = 1$; the congruence $h\bar{h} \equiv -1 \pmod{2}$ gives $\bar{h} \equiv 1$. The exponential is $e^{2\pi i(n \cdot 1 + (-1) \cdot 1)/2} = e^{\pi i(n-1)} = (-1)^{n-1}$. The smallest cases, worked by hand: $n = 1$ gives $K_2(1, -1; 2) = \bar{\nu}(\gamma_{1,2})$, and $n = 2$ gives $K_2(2, -1; 2) = -\bar{\nu}(\gamma_{1,2})$. $\square$

Proposition 4.6 (Characterisation of the alternation). *The identity*

$$K_2(n, -1; 2) = (-1)^n \quad \text{for all } n \geq 1 \quad (5)$$

holds if and only if $\bar{\nu}(\gamma_{1,2}) = -1$. For the trivial multiplier — the classical sum — the value is $(-1)^{n-1} = -(-1)^n$, and (5) fails at every n .

Proof. By Lemma 4.5, $K_2(n, -1; 2) = \bar{\nu}(\gamma_{1,2})(-1)^{n-1}$. If $\bar{\nu}(\gamma_{1,2}) = -1$ then $K_2(n, -1; 2) = -(-1)^{n-1} = (-1)^n$ for every n ; conversely, evaluating (5) at $n = 1$ gives $\bar{\nu}(\gamma_{1,2}) = -1$. Setting $\bar{\nu} \equiv 1$ in (4) gives the classical evaluation. Both the characterisation and the classical contrast are machine-checked (Appendix A, Certificate 8). $\square$

Remark 4.7. We do not evaluate $\bar{\nu}(\gamma_{1,2})$ here. Proposition 4.6 states exactly what the parity of Proposition 4.3 determines and what it does not: the transit parity $(-1)^n$ coincides with the value of the multiplier-weighted sum (3) at modulus two for all n precisely when the single multiplier value is -1 , and we identify the two under that condition, as an identification rather than a theorem. The contrast with the classical sum shows that the identification genuinely selects the multiplier-weighted object: the two differ by an overall sign at every n .

The relation of this section to the class of Section 6 is one of structure, not derivation, and is stated with the class hypotheses. Here we record only the observation that the parity of Proposition 4.3 and the counting function d_{odd} of Section 3 are attached to the same level-two object: the sums (3) accompany the coefficients of Appell-type series, of which the level-two sum (2) is the member at hand, in expansions of Rademacher type [3], so that the two are faces of a single arithmetic structure rather than independent inputs.

5 Framed moduli and the vanishing of odd cohomology

This section proves the homological statement and assembles the generating-function context that connects the moduli invariants to the coefficients of Section 3. Throughout, $M(r, n)$ denotes the framed moduli space of torsion-free sheaves E on the projective plane $\mathbb{P}^2$ with rank r and $c_2(E) = n$, the framing being a trivialisation of the restriction of E to the line at infinity ℓ_∞ ; these are the completions of the moduli of framed instantons on $\mathbb{C}^2 \cong \mathbb{R}^4$, and they are smooth quasi-projective varieties of dimension $2rn$ [9].

Theorem 5.1 (Vanishing of odd cohomology). *The framed moduli space $M(r, n)$ admits a decomposition into affine cells of even real dimension. Consequently $H_{2j+1}(M(r, n), \mathbb{Z}) = 0$ for all j , and the Euler characteristic equals the absolute Betti count:*

$$\chi(M(r, n)) = \sum_j (-1)^j b_j = \sum_j b_{2j} = \sum_j b_j. \quad (6)$$

There are, in particular, no cancellations: the counting invariant $\chi(M(r, n))$ is an absolute count of cells.

Proof. The maximal torus of $GL_r(\mathbb{C})$ acting on the framing, together with the two-dimensional torus acting on $\mathbb{P}^2$ preserving ℓ_∞ , acts on $M(r, n)$ with finitely many fixed points, parametrised by r -tuples of Young diagrams with n boxes in total [9]. For a generic one-parameter subgroup of this torus, the Białynicki-Birula theorem [1] decomposes $M(r, n)$ into locally closed attracting cells, one for each fixed point, each cell being an affine space; the existence of the limits required by the decomposition in the quasi-projective setting is established for these moduli spaces in [9] (for $r = 1$ the statement is the cell decomposition of the Hilbert schemes of points [8]; the resulting Betti numbers of $M(r, n)$ are computed explicitly in [10]). Affine cells have even real dimension, so the cellular chain complex is concentrated in even degrees, and all odd Betti numbers vanish. The identity (6) is then term-by-term: for even j the sign is $+1$, and for odd j the Betti number is zero. The parity implication — vanishing odd part gives signed sum equal to plain sum, for any Betti sequence — is machine-checked in full generality in Appendix A (Certificate 6). $\square$

Theorem 5.1 is the statement that the counting invariants of the framed moduli spaces are counts without cancellation. Two classical generating identities organise these counts. First, for $r = 1$ the moduli space $M(1, n)$ is isomorphic to the Hilbert scheme of n points on $\mathbb{C}^2$ [8], and for a smooth projective surface S Göttsche's formula [6] gives

$$\sum_{n \geq 0} \chi(\text{Hilb}^n(S)) q^n = \prod_{m \geq 1} (1 - q^m)^{-\chi(S)}. \quad (7)$$

The smallest case, worked by hand: $\text{Hilb}^1(S) = S$, and the coefficient of q^1 on the right of (7) is $\chi(S)$, as it must be. Second, for higher rank on the projective plane, the partition functions assembled from the invariants of $M(r, n)$ in the topologically twisted theory of Vafa and Witten [13] are built from generalised Appell functions: this is proved by Bringmann, Manschot and Rolin [2], whose identities express higher-rank counting series on the projective plane through generalised Appell functions, the family of which the level-two sum (2) is the depth-one member. The central coefficients of that member are, by Proposition 3.4, exactly $d_{\text{odd}}(n)$. The counting object to which the class hypotheses of Section 6 refer is therefore assembled entirely from ingredients each of which stands on the results of this section and of Section 3: cell counts without cancellation (Theorem 5.1), organised by Appell-type series whose central coefficients are $d_{\text{odd}}(n)$ (Proposition 3.4).

6 The theory class

We now define the class of theories for which the gap of Section 2 will be identified with a mass. The definition is by explicit spectral hypotheses; the class carries the entire conditional weight of the paper, and nothing in this section is proved about membership of the class.

Definition 6.1 (The class $\mathcal{C}$). A *Kaluza–Klein-reduced theory over the setting of Definition 2.1* is a four-dimensional theory $\mathcal{T}$ obtained by reduction on a compact internal space containing the pillowcase fibre, presented through its one-particle spectral data: a degeneracy function $m: \mathbb{N} \rightarrow \mathbb{N}$, a mass assignment $\text{mass}: \mathbb{N} \rightarrow \mathbb{R}$, and the set $\mathcal{S}(\mathcal{T}) \subset \mathbb{R}$ of masses of non-vacuum one-particle states. The theory $\mathcal{T}$ belongs to the class $\mathcal{C}$ if:

- (A1) the level degeneracies satisfy $m(n) = d_{\text{odd}}(n)$ for all $n \geq 1$;
- (A2) the mass at level n equals $n\Delta$, where Δ is the corner gap (1) of the internal setting;
- (A3) $d_{\text{odd}}(n)$ counts the one-particle states of $\mathcal{T}$ at mass level n ; equivalently, $\mathcal{S}(\mathcal{T})$ consists exactly of the values $\text{mass}(n)$ over the occupied levels $n \geq 1$, those with $m(n) \neq 0$.

Three comments delimit what Definition 6.1 does and does not assert. First, the hypotheses (A1)–(A3) are declared, not derived: whether a given reduction satisfies them is a question about that reduction, outside the scope of this paper. The parity computation of Section 4 does not derive hypothesis (A1); it exhibits the arithmetic structure that (A1) posits as the orbit signature of an elementary group action, so that the hypothesis is one about a quantity with independent algebraic meaning. Second, hypothesis (A3) is part of the definition of the class and is not a consequence of Theorem 5.1. What Theorem 5.1 supplies is the consistency of the reading: the counting coefficients to which (A3) refers are absolute counts, free of cancellation, so that a hypothesis identifying them with numbers of states is a hypothesis about unsigned quantities. The identification itself remains an assumption. Third, the deflationary reading of the main results is therefore the correct one: Section 7 proves exact inferences from (A1)–(A3), and the strength of the conclusions for any particular theory is precisely the strength of the case that the theory satisfies the hypotheses.

Remark 6.2 (A second route to the absence of a paired sector). Hypothesis (A3) excludes, by fiat, a paired sector: states cancelling in a signed count. Theorem 5.1 removes such a sector at the homological level. It is worth recording that a modular argument reaches the same exclusion from a different hypothesis. Suppose the completed counting series of a theory — the series whose n -th coefficient is the number of states at level n , completed so as to transform as a modular form of weight zero without poles — differs from the index series by the generating function of a putative paired sector, $\sum_n 2k_n q^n$. A holomorphic modular form of weight zero without poles is constant — for the full modular group this is classical [12, Ch. VII], and in general it is the statement that a holomorphic function on a compact Riemann surface is constant — and since the vacuum is unpaired, $k_0 = 0$, the constant is zero, so $k_n = 0$ identically. Weight zero is the natural home for the hypothesis: the difference of two counting series for the same graded object, with the same normalisation, carries no automorphy weight of its own, and constancy is then the full strength of the Liouville-type statement. The remark is in this sense an arithmetic mirror of Theorem 5.1: the homological argument excludes a paired sector at the level of cells, the modular argument excludes it at the level of completed series, and neither mechanism implies the other. The hypothesis of modular completion is not part of the definition of $\mathcal{C}$ and is not used elsewhere in this paper; the remark records only that two independent mechanisms both close the paired-sector loophole under their respective hypotheses.

7 The spectral gap and its gauge-group independence

We now prove the two conditional results. Throughout, $\mathcal{T} \in \mathcal{C}$ is a theory in the class of Definition 6.1, over a fixed setting (M, g, ι, c) , and Δ is the corner gap (1).

Proposition 7.1 (Identification of Δ with the lowest mass). *Let $\mathcal{T} \in \mathcal{C}$. Then Δ is the mass of the lowest non-vacuum one-particle state of $\mathcal{T}$; that is, Δ is the least element of $\mathcal{S}(\mathcal{T})$.*

Proof. By (A3) with (A1), the non-vacuum one-particle masses of $\mathcal{T}$ are exactly the values $\text{mass}(n)$ over the levels $n \geq 1$ with $d_{\text{odd}}(n) \neq 0$. The smallest case, worked by hand: $d_{\text{odd}}(1) = 1$, the sole divisor of 1 being odd, so level $n = 1$ is occupied. By (A2), $\text{mass}(n) = n\Delta$; and by Lemma 2.3, $\Delta \geq 0$, so $n\Delta \geq \Delta$ for all $n \geq 1$, with equality at $n = 1$. Hence Δ belongs to $\mathcal{S}(\mathcal{T})$ and bounds it below.

We record explicitly that the proof consumes the three declared hypotheses together with one internal input, $\Delta \geq 0$, which is discharged by Lemma 2.3 and is not an additional assumption. The formal certificate (Appendix A, Certificate 3) carries this input as an explicit argument, discharged from the geometry in Certificate 4, so that the dependence of the identification on the positivity lemma is visible rather than silent. $\square$

Proposition 7.2 (Gauge-group independence of the corner gap). *Let $\mathcal{T} \in \mathcal{C}$ have gauge group G . Then:*

- (i) $\Delta > 0$;
- (ii) Δ is the mass of the lowest non-vacuum one-particle state of $\mathcal{T}$;
- (iii) neither (i) nor (ii) refers to G .

Consequently the existence of a positive mass gap holds for every theory in $\mathcal{C}$, uniformly in the gauge group.

Proof. Claim (i) is Lemma 2.3, whose sole geometric input is Lemma 2.2. Claim (ii) is Proposition 7.1 under the hypotheses (A1)–(A3) of Definition 6.1. Claim (iii) is an inspection

of dependencies. The complete input set of (i) and (ii) consists of five items: the two-torsion structure of σ on $\mathbb{T}^2$ (Lemma 2.2); the Hopf–Rinow theorem (Lemma 2.3); and the three hypotheses (A1), (A2), (A3). None of these five inputs mentions G : the first two are statements about the internal geometry, and the three hypotheses constrain spectral data in which the gauge group does not appear. The gauge group of $\mathcal{T}$ is whatever the reduction produces; it enters the discussion only when a particular theory is named, and it enters downstream of, not upstream of, (i) and (ii).

The inspection is verified mechanically. In the formal development (Appendix A, Certificate 4), the master theorem is quantified over an arbitrary type G , and G occurs in none of the hypotheses, definitions, or conclusions; the proof term is constant in G , witnessed by the fact that the identity of proofs at two distinct instantiations is definitional. This is the machine-checked form of claim (iii). $\square$

8 Scope and outlook

Corollary 8.1 (Scope: the value of Δ is not gauge-group independent). *Proposition 7.2 establishes the gauge-group independence of the existence claim $\Delta > 0$ only. The numerical value of Δ depends on which corner is distinguished and on the induced metric data — the selection (g, ι, c) of Definition 2.1 — and is therefore not a gauge-group invariant; no such claim is made here.*

Proof. The existence half is uniform: positivity consumes only distinctness (Lemma 2.2) and the metric-space structure (Lemma 2.3), and these hold for every admissible selection. The value half is witnessed by exhibiting two admissible selections with different gap values; an explicit pair, with gaps in the ratio two to one, is constructed in the formal development (Appendix A, Certificate 5). Hence the value is a non-constant function of the selection data and cannot be an invariant of anything that does not determine the selection. $\square$

The two halves of Corollary 8.1 have different logical characters, and the distinction is load-bearing. Existence is uniform over selections and over gauge groups; value is a function of the selection. A reader interested in a particular reduced theory should therefore expect the present results to supply the existence of its gap, conditional on membership of $\mathcal{C}$, and to be silent on the gap’s magnitude. Which corner is distinguished is likewise data from the present standpoint; we note only that in reductions whose internal space carries structure beyond the metric — an orientation of the fibre, or a choice of spin or line-bundle data — the selection can be induced by that structure rather than left free, and the value of Δ is then determined by the same data that determine the theory.

Two directions merit note. First, the class $\mathcal{C}$ is not empty as defined: a spectral-data triple with $m(n) = d_{\text{odd}}(n)$ and $\text{mass}(n) = n\Delta$, over any admissible setting of Definition 2.1, satisfies (A1)–(A3) by construction, and the formal development of Appendix A instantiates exactly such data. The open question is therefore not non-emptiness but dynamical realisation: whether a reduction of physical interest produces this spectral data from its geometry. Exhibiting one, with the pillowcase realised as an orbifold fibre of the internal space and the degeneracies computed from its geometry, would convert the conditional results into statements about that theory. The natural shape of such a candidate is visible in the structure of the count itself: we observe that anti-periodic boundary conditions on an order-two quotient produce half-integer moding, that is, spectra supported on odd multiples of a basic spacing, and $d_{\text{odd}}(m) = \#\{(n, a) : n(2a + 1) = m\}$ is precisely the enumeration of a level m by such odd multiples (Proposition 3.4); the generating-function context of Section 5 indicates where degeneracies of this form arise. Second, the structure of Proposition 7.2 — a gap proved from fixed-point topology and metric completeness, with the gauge group entering nowhere

— suggests that existence questions for gaps may in general be separable from the dynamical questions that determine values. An existence-only result of this kind may be useful in making progress towards the mass-gap problem for four-dimensional gauge theories, where it is the existence, not the value, that is asked for.

Acknowledgements

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A Formal certificates

Eight machine-checked certificates accompany the paper, developed in Lean 4 (toolchain 4.15.0) against Mathlib (tag v4.15.0). The suite compiles with zero errors, zero warnings, and no incomplete proofs; an axiom audit confirms that every headline theorem depends only on the three standard axioms of the Lean/Mathlib foundation (propositional extensionality, choice, and quotient soundness). Certificates 1 and 2 are unconditional and cover Lemmas 2.2 and 2.3, the latter at metric-space level as noted in Section 2. Certificate 3 is conditional and machine-checks Proposition 7.1: the hypotheses (A1)–(A3), and exactly those, are encoded as named fields, with the internal input $\Delta \geq 0$ an explicit argument. Certificate 4 machine-checks Proposition 7.2, with the gauge group a phantom type parameter and the constancy of the proof term in that parameter witnessed definitionally. Certificate 5 is the statement-level form of Corollary 8.1, including the explicit two-selection witness. Certificate 6 proves the parity implication of Theorem 5.1 in full generality, and Certificate 7 proves the coefficient identity of Proposition 3.4 for all $m \geq 1$ via the explicit divisor bijection, together with the certified table to order fifteen. Certificate 8 has three parts: an unconditional part machine-checking the orbit decomposition of Lemma 4.2 and the convention-independence of the sign, by exhaustive decision on an explicit eight-element model; a conditional part machine-checking the inference of Proposition 4.3, with the two lift identifications encoded as named hypothesis fields in the pattern of Certificate 3; and a second unconditional part machine-checking the characterisation of Proposition 4.6 together with the classical-sum contrast.

Certificate 1 — corner distinctness (`Corners.lean`)

```

/-
Certificate 1 (Lemma: corner distinctness).
On the torus  $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ , the involution  $\sigma : (x,y) \mapsto (-x,-y)$  has
exactly four fixed points -- the 2-torsion points
 $(0,0), (1/2,0), (0,1/2), (1/2,1/2)$  -- and they are pairwise distinct.
Unconditional; zero `sorry`. No reference to any gauge group, and no
quaternionic structure, appears in this file.
Model: the circle is Mathlib's AddCircle (1 : ℝ) = ℝ/ℤ; the torus
is the product of two copies -- the object of the prose statement
itself, not a substitute model.
-/
import Mathlib.Topology.Instances.AddCircle

noncomputable section
namespace Corners

open AddCircle

/-- The circle  $\mathbb{R}/\mathbb{Z}$ . -/
abbrev S : Type := AddCircle (1 : ℝ)

```

```

/-- The torus  $T^2 = (\mathbb{R}/\mathbb{Z})^2$ . -/
abbrev T : Type := S × S

/-- The involution  $\sigma(x,y) = (-x,-y)$ . -/
def sigma (q : T) : T := (-q.1, -q.2)

/-- The four corner points. -/
def c00 : T := (((0 : ℝ) : S), ((0 : ℝ) : S))
def c10 : T := (((1 : ℝ)/2 : ℝ) : S), ((0 : ℝ) : S))
def c01 : T := (((0 : ℝ) : S), ((1 : ℝ)/2 : ℝ) : S))
def c11 : T := (((1 : ℝ)/2 : ℝ) : S), ((1 : ℝ)/2 : ℝ) : S))

/-! ### Single-circle facts -/

/-- An integer is zero on the circle. -/
theorem int_coe_eq_zero (m : ℤ) : (((m : ℝ)) : S) = ((0 : ℝ) : S) := by
  have h : (((m : ℝ)) : S) = 0 :=
    (AddCircle.coe_eq_zero_iff (1 : ℝ)).mpr ⟨m, by rw [zsmul_eq_mul, mul_one]⟩
  rw [h, show ((0 : ℝ) : S) = 0 by simp]

/-- 1/2 is not zero on the circle. -/
theorem half_ne_zero : (((1 : ℝ)/2 : ℝ) : S) ≠ ((0 : ℝ) : S) := by
  intro h
  have h0 : (((1 : ℝ)/2 : ℝ) : S) = 0 := by simpa using h
  rw [AddCircle.coe_eq_zero_iff] at h0
  obtain ⟨n, hn⟩ := h0
  rw [zsmul_eq_mul, mul_one] at hn
  have h2 : ((2 * n : ℤ) : ℝ) = 1 := by push_cast; linarith
  have h3 : (2 * n : ℤ) = 1 := by exact_mod_cast h2
  omega

/-- A point of the circle is fixed by negation iff it is 0 or 1/2. -/
theorem neg_eq_self_iff (x : S) :
  -x = x ↔ x = ((0 : ℝ) : S) ∨ x = (((1 : ℝ)/2 : ℝ) : S) := by
  refine QuotientAddGroup.induction_on x (fun r => ?_)
  constructor
  · intro h
    have hneg : ((-r : ℝ) : S) = ((r : ℝ) : S) := by
      rw [AddCircle.coe_neg]; exact h
    have h0 : ((r + r : ℝ) : S) = 0 := by
      have := sub_eq_zero.mpr hneg.symm
      rw [← AddCircle.coe_sub] at this
      simpa [sub_neg_eq_add] using this
    rw [AddCircle.coe_eq_zero_iff] at h0
    obtain ⟨n, hn⟩ := h0
    rw [zsmul_eq_mul, mul_one] at hn
    rcases Int.even_or_odd n with ⟨m, hm⟩ | ⟨m, hm⟩
    · left
      have hr : r = (m : ℝ) := by
        have : ((m : ℝ) + (m : ℝ)) = r + r := by
          rw [← hn, hm]; push_cast; ring
          linarith
      rw [hr]; exact int_coe_eq_zero m
    · right
      have hr : r = (m : ℝ) + 1/2 := by
        have : ((2 * m + 1 : ℤ) : ℝ) = r + r := by rw [← hn, hm]
        push_cast at this; linarith
      have : ((r - 1/2 : ℝ) : S) = ((0 : ℝ) : S) := by
        rw [show (r - 1/2 : ℝ) = (m : ℝ) by rw [hr]; ring]
        exact int_coe_eq_zero m
      have h5 : ((r : ℝ) : S) - (((1:ℝ)/2 : ℝ) : S) = 0 := by

```

```

      rw [← AddCircle.coe_sub]; simp using this
      exact sub_eq_zero.mp h5
    · rintro (h | h)
      · rw [h]; simp
      · rw [h]
      have : ((-((1:ℝ)/2) : ℝ) : S) = (((1:ℝ)/2 : ℝ) : S) := by
        have : ((-((1:ℝ)/2) - (1:ℝ)/2 : ℝ) : S) = ((0 : ℝ) : S) := by
          rw [show (-((1:ℝ)/2) - (1:ℝ)/2 : ℝ) = ((-1 : ℤ) : ℝ) by push_cast; ring]
          exact int_coe_eq_zero (-1)
        have h5 : ((-((1:ℝ)/2) : ℝ) : S) - (((1:ℝ)/2 : ℝ) : S) = 0 := by
          rw [← AddCircle.coe_sub]; simp using this
          exact sub_eq_zero.mp h5
      rw [← AddCircle.coe_neg] at *
      exact this

/-! ### The torus theorems -/

/-- **Classification.** The fixed-point set of  $\sigma$  is exactly the four
corner points. -/
theorem fixed_iff (q : T) :
  sigma q = q ↔ q = c00 ∨ q = c10 ∨ q = c01 ∨ q = c11 := by
  obtain ⟨x, y⟩ := q
  have hx := neg_eq_self_iff x
  have hy := neg_eq_self_iff y
  simp only [sigma, Prod.mk.injEq, c00, c10, c01, c11]
  constructor
  · rintro (h1, h2)
    rcases hx.mp h1 with rfl | rfl <|> rcases hy.mp h2 with rfl | rfl <|> tauto
  · rintro ((rfl, rfl) | (rfl, rfl) | (rfl, rfl) | (rfl, rfl)) <|>
    exact (hx.mpr (by tauto), hy.mpr (by tauto))

/-- Each corner is fixed. -/
theorem corners_fixed :
  sigma c00 = c00 ∧ sigma c10 = c10 ∧ sigma c01 = c01 ∧ sigma c11 = c11 :=
  ((fixed_iff c00).mpr (by tauto), (fixed_iff c10).mpr (by tauto),
  (fixed_iff c01).mpr (by tauto), (fixed_iff c11).mpr (by tauto))

/-- **Distinctness.** The four corners are pairwise distinct. -/
theorem corners_pairwise_distinct :
  c00 ≠ c10 ∧ c00 ≠ c01 ∧ c00 ≠ c11 ∧
  c10 ≠ c01 ∧ c10 ≠ c11 ∧ c01 ≠ c11 :=
  (fun h => half_ne_zero (congrArg Prod.fst h).symm,
  fun h => half_ne_zero (congrArg Prod.snd h).symm,
  fun h => half_ne_zero (congrArg Prod.fst h).symm,
  fun h => half_ne_zero (congrArg Prod.fst h),
  fun h => half_ne_zero (congrArg Prod.snd h).symm,
  fun h => half_ne_zero (congrArg Prod.fst h).symm)

/-- **Summary.**  $\sigma$  has exactly the four corners as fixed points, and
they are pairwise distinct. -/
theorem corner_distinctness :
  (∀ q : T, sigma q = q ↔ q = c00 ∨ q = c10 ∨ q = c01 ∨ q = c11) ∧
  (c00 ≠ c10 ∧ c00 ≠ c01 ∧ c00 ≠ c11 ∧
  c10 ≠ c01 ∧ c10 ≠ c11 ∧ c01 ≠ c11) :=
  (fixed_iff, corners_pairwise_distinct)

end Corners

```

Certificate 2 — positivity of the corner gap (Gap.lean)

```

/-
Certificate 2 (Lemma: positivity of the corner gap).

```

Distinct points of a metric space are at strictly positive distance;
 instantiated at the four corner points of Certificate 1, giving $\Delta > 0$.
 Unconditional; zero `sorry`.

Formalisation note (stated once): the certificate is proved at
 METRIC-SPACE level -- strictly weaker machinery than Hopf--Rinow and
 strictly sufficient for the claim as used. The prose retains
 Hopf--Rinow as the Riemannian-geometric source: the ambient manifold
 M, compact with continuous Riemannian metric, induces a metric-space
 structure, and `ι` below is the injective inclusion of the pillowcase
 fibre carrying the induced distance.

No reference to any gauge group appears in this file.

-/

import YML.Corners

noncomputable section

namespace Gap

open Corners

/-- Distinct points of a metric space are at strictly positive
 distance. -/

theorem dist_pos_of_ne {X : Type*} [MetricSpace X] {a b : X}
 (h : a ≠ b) : 0 < dist a b :=
 dist_pos.mpr h

/-- The minimum of three positive reals is positive. -/

theorem min₃_pos {a b c : ℝ} (ha : 0 < a) (hb : 0 < b) (hc : 0 < c) :
 0 < min a (min b c) :=
 lt_min ha (lt_min hb hc)

section Embedding

variable {K : Type*} [MetricSpace K]

variable (ι : T → K)

/-- All six pairwise corner distances in `K` are positive. -/

theorem corner_dists_pos (hι : Function.Injective ι) :
 0 < dist (ι c00) (ι c10) ∧ 0 < dist (ι c00) (ι c01) ∧
 0 < dist (ι c00) (ι c11) ∧ 0 < dist (ι c10) (ι c01) ∧
 0 < dist (ι c10) (ι c11) ∧ 0 < dist (ι c01) (ι c11) := by
 obtain ⟨h1, h2, h3, h4, h5, h6⟩ := corners_pairwise_distinct
 exact (dist_pos_of_ne (fun h => h1 (hι h)),
 dist_pos_of_ne (fun h => h2 (hι h)),
 dist_pos_of_ne (fun h => h3 (hι h)),
 dist_pos_of_ne (fun h => h4 (hι h)),
 dist_pos_of_ne (fun h => h5 (hι h)),
 dist_pos_of_ne (fun h => h6 (hι h)))

/-- **Positivity of the corner gap.** The minimum distance from the
 distinguished corner to the three remaining corners is strictly
 positive. Stated with `c00` as the distinguished exemplar; the
 argument is corner-symmetric, and the selection data pick which
 corner plays the role. -/

theorem gap_pos (hι : Function.Injective ι) :
 0 < min (dist (ι c00) (ι c10))
 (min (dist (ι c00) (ι c01)) (dist (ι c00) (ι c11))) := by
 obtain ⟨h1, h2, h3, _, _, _⟩ := corner_dists_pos ι hι
 exact min₃_pos h1 h2 h3

end Embedding

end Gap

Certificate 3 — conditional identification (Identification.lean)

```

/-
Certificate 3 (Proposition: conditional identification).
 $\Delta$  is the mass of the lowest non-vacuum one-particle state of a theory
in the class C. CONDITIONAL: the certificate machine-checks the
conditional structure -- the three class hypotheses (A1)-(A3), and
exactly those three, are encoded as named fields, and the conclusion
is derived from them. It does not prove the hypotheses.
  hA1 -- (A1): level degeneracies  $m(n) = d_{\text{odd}}(n)$ 
  hA2 -- (A2): the mass at level  $n$  equals  $n \cdot \Delta$ 
  hA3 -- (A3):  $d_{\text{odd}}(n)$  counts one-particle states at level  $n$ , so the
        non-vacuum one-particle masses are exactly the occupied
        levels  $n \geq 1$ 
 $d_{\text{odd}}$  itself is NOT a hypothesis: it is defined arithmetically (count
of odd divisors) and its needed value  $d_{\text{odd}}(1) = 1$  is proved.
FINDING (made visible rather than silent): the lower-bound direction
needs  $0 \leq \Delta$ . In the prose this is supplied by the positivity lemma;
here it is an explicit argument `hApos`, and the master certificate
(Certificate 4) discharges it from the geometry.
No reference to any gauge group appears in this file.
-/
import Mathlib.NumberTheory.Divisors
import Mathlib.Analysis.SpecialFunctions.Pow.Real

noncomputable section
namespace Identification

/-- The odd-divisor count  $d_{\text{odd}}(n)$ , defined arithmetically. -/
def dodd (n : ℕ) : ℕ := ((Nat.divisors n).filter (fun d => Odd d)).card

/-- The smallest case, worked by machine:  $d_{\text{odd}}(1) = 1$ . -/
theorem dodd_one : dodd 1 = 1 := by decide

/-- The identification hypotheses, exactly the three of the class. -/
structure IdentificationData where
  /-- the internal spectral separation between the distinguished
       corner and the nearest remaining corner (geometric input;
       positive by the gap lemma when instantiated -- assumed here,
       discharged in Certificate 4) -/
   $\Delta$  : ℝ
  /-- level degeneracies of the reduced theory -/
  m : ℕ → ℕ
  /-- (A1):  $m(n) = d_{\text{odd}}(n)$  -/
  hA1 : ∀ n, m n = dodd n
  /-- one-particle mass assignment at level  $n$  -/
  mass : ℕ → ℝ
  /-- (A2): the activated spectrum is  $\Delta$ -spaced -/
  hA2 : ∀ n, mass n = n *  $\Delta$ 
  /-- the set of masses of non-vacuum one-particle states -/
  massSet : Set ℝ
  /-- (A3): the non-vacuum one-particle masses are exactly the
       occupied levels  $n \geq 1$  -/
  hA3 : massSet = { x | ∃ n : ℕ, 1 ≤ n ∧ m n ≠ 0 ∧ x = mass n }

/-- **Conditional identification.** Under the three hypotheses,  $\Delta$  is
the least element of the set of non-vacuum one-particle masses. -/
theorem identification (D : IdentificationData) (hApos : 0 ≤ D. $\Delta$ ) :
  IsLeast D.massSet D. $\Delta$  := by
  constructor
  · -- membership: level 1 is occupied ( $d_{\text{odd}}(1) = 1$ ) and has mass  $\Delta$ 
    rw [D.hA3]
    refine (1, le_refl 1, ?_, ?_)

```

```

    · rw [D.hA1, dodd_one]; exact one_ne_zero
    · rw [D.hA2]; simp
  · -- lower bound: every occupied level  $n \geq 1$  has mass  $n\Delta \geq \Delta$ 
    rintro x hx
    rw [D.hA3] at hx
    obtain ⟨n, h1, _, rfl⟩ := hx
    rw [D.hA2]
    calc D.Δ = 1 * D.Δ := (one_mul _).symm
    _ ≤ n * D.Δ := by
      apply mul_le_mul_of_nonneg_right _ hΔpos
      exact_mod_cast h1
end Identification

```

Certificate 4 — gauge transfer (Transfer.lean)

```

/-
Certificate 4 (Proposition: gauge transfer, master form).
The transfer proposition assembled, with claim (iii) -- gauge-group
independence -- made STRUCTURAL. The master theorem is quantified over
an arbitrary type `G` ("the gauge group"), and `G` occurs in NONE of
the hypotheses, definitions, or conclusions: the proof term is
constant in `G` (verified by `rfl`). This is the machine-checkable
form of the inspection of the proof's dependencies: the five inputs --
  (1) the 2-torsion structure of  $\sigma$  on  $T^2$  [Certificate 1, proved]
  (2) positive distance between distinct points [Certificate 2, proved]
  (3) (A1), (4) (A2), (5) (A3) [Certificate 3, named]
-- are all G-free, verified by Lean's binder discipline.
Conditional on (A1)-(A3); unconditional in G.
-/
import YML.Corners
import YML.Gap
import YML.Identification

noncomputable section
namespace Transfer
open Corners Gap Identification

section Master

variable {K : Type*} [MetricSpace K] -- the ambient manifold, as a metric space
variable (ι : T → K) -- inclusion of the pillowcase fibre

/-- The corner gap: distance from the distinguished corner to the
nearest remaining corner. `c00` as the distinguished exemplar; the
argument is corner-symmetric. -/
def cornerGap : ℝ :=
  min (dist (ι c00) (ι c10)) (min (dist (ι c00) (ι c01)) (dist (ι c00) (ι c11)))

-- NOTE: without the option below, Lean's linter reports
-- "unused variable `G`" on the theorem that follows. That report IS
-- claim (iii), issued by the machine itself. It is silenced only to
-- keep the compile warning-free, having been placed on the record.
set_option linter.unusedVariables false in
/- **Gauge transfer (master form).** For an ARBITRARY gauge group
`G` -- which occurs nowhere in the hypotheses -- if the fibre embeds
injectively in `K` and the identification data carries  $\Delta = \text{cornerGap}$ ,
then (i)  $\Delta > 0$ , and (ii)  $\Delta$  is the mass of the lowest non-vacuum
one-particle state. Claim (iii) is the visible fact that `G` is a
phantom parameter. -/
theorem gauge_transfer (G : Type*)
  (hι : Function.Injective ι)

```



```

(D : IdentificationData)
(hΔdef : D.Δ = cornerGap ι) :
  0 < D.Δ ∧ IsLeast D.massSet D.Δ := by
have hpos : 0 < D.Δ := by rw [hΔdef]; exact gap_pos ι h1
exact (hpos, identification D (le_of_lt hpos))

/-- The proof term is literally constant in `G`. -/
theorem transfer_constant_in_G (G1 G2 : Type*)
  (hι : Function.Injective ι) (D : IdentificationData)
  (hΔdef : D.Δ = cornerGap ι) :
  gauge_transfer ι G1 h1 D hΔdef = gauge_transfer ι G2 h1 D hΔdef :=
  rfl

end Master
end Transfer

```

Certificate 5 — scope (Scope.lean)

```

/-
Certificate 5 (Corollary: scope, statement-level).
The EXISTENCE claim  $\Delta > 0$  is gauge-group independent; the numerical
VALUE of  $\Delta$  is not: it depends on the corner-selection data, and no
value-independence claim is made.
Formalisation (statement-level): a "selection" is the data the value
actually depends on -- the distinguished corner image and the three
remaining corner images, pairwise distinct from it. Two theorems:
  (1) existence is selection- and G-uniform;
  (2) two explicit admissible selections with DIFFERENT gap values
      (1 and 2), witnessing that the value is a function of the
      selection, not a gauge-group invariant.
As in Certificate 4, `G` is a phantom parameter throughout.
Unconditional (statement-level); zero `sorry`.
-/
import YML.Gap

noncomputable section
namespace Scope
open Gap

/-- A corner selection in a metric space `K`: the distinguished corner
image and the three remaining corner images, pairwise distinct from
it. (The data the value of  $\Delta$  actually depends on.) -/
structure Selection (K : Type*) [MetricSpace K] where
  vac : K
  o1 : K
  o2 : K
  o3 : K
  h1 : vac ≠ o1
  h2 : vac ≠ o2
  h3 : vac ≠ o3

/-- The gap of a selection. -/
def Selection.gap {K : Type*} [MetricSpace K] (s : Selection K) : ℝ :=
  min (dist s.vac s.o1) (min (dist s.vac s.o2) (dist s.vac s.o3))

-- The linter's "unused variable `G`" report is the scope claim's
-- gauge-independence half, issued by the machine; silenced only to
-- keep the compile warning-free, having been placed on the record.
set_option linter.unusedVariables false in
/-
**Existence half.** For every admissible selection and every
gauge group `G`, the gap is positive. Selection- and G-uniform. -/
theorem existence_selection_and_gauge_independent

```

```

    (G : Type*) {K : Type*} [MetricSpace K] (s : Selection K) :
    0 < s.gap :=
    min₃_pos (dist_pos_of_ne s.h1) (dist_pos_of_ne s.h2) (dist_pos_of_ne s.h3)

/-! ### Value half: two admissible selections, different values -/

/-- Selection A: corners at 0, 1, 2, 3 on the real line. -/
def selA : Selection ℝ where
  vac := 0
  o1 := 1
  o2 := 2
  o3 := 3
  h1 := by norm_num
  h2 := by norm_num
  h3 := by norm_num

/-- Selection B: corners at 0, 2, 4, 6 on the real line. -/
def selB : Selection ℝ where
  vac := 0
  o1 := 2
  o2 := 4
  o3 := 6
  h1 := by norm_num
  h2 := by norm_num
  h3 := by norm_num

theorem gapA : selA.gap = 1 := by
  simp only [Selection.gap, selA, Real.dist_eq]
  norm_num

theorem gapB : selB.gap = 2 := by
  simp only [Selection.gap, selB, Real.dist_eq]
  norm_num

/-- **Value half.** The value of the gap is not
selection-independent: two admissible selections with different
values. Hence no value-independence claim can be made. -/
theorem value_depends_on_selection : selA.gap ≠ selB.gap := by
  rw [gapA, gapB]; norm_num

end Scope

```

Certificate 6 — signed count equals absolute count (BettiParity.lean)

```

/-
Certificate 6 (Theorem: signed count equals absolute count).
The arithmetic implication of the index theorem: if all odd Betti
numbers vanish, the signed Euler characteristic equals the absolute
Betti count -- there are no cancellations. The geometric inputs (the
cellular decomposition and the vanishing itself) are external
theorems cited in the prose; this certificate machine-checks the
implication, in full generality (any Betti sequence, any range),
not merely at statement level.
Unconditional; zero `sorry`.
-/
import Mathlib.Algebra.BigOperators.Group.Finset
import Mathlib.Tactic.NormNum
import Mathlib.Algebra.Order.BigOperators.Group.Finset

namespace BettiParity

open Finset

```

```

/-- **Signed = absolute.** For any Betti sequence  $b : \mathbb{N} \rightarrow \mathbb{Z}$  with
vanishing odd part, the alternating sum equals the plain sum over any
initial range. -/
theorem signed_eq_absolute (b :  $\mathbb{N} \rightarrow \mathbb{Z}$ )
  (hodd :  $\forall j, b (2 * j + 1) = 0$ ) (N :  $\mathbb{N}$ ) :
   $\sum j \in \text{range } N, (-1 : \mathbb{Z}) ^ j * b j = \sum j \in \text{range } N, b j :=$  by
  apply Finset.sum_congr rfl
  intro j _
  rcases Nat.even_or_odd j with ⟨k, hk⟩ | ⟨k, hk⟩
  · -- even j: the sign is +1
    subst hk
    rw [show k + k = 2 * k by omega, pow_mul]
    norm_num
  · -- odd j: the term vanishes on both sides
    subst hk
    rw [hodd k, mul_zero]

/-- Corollary in the form used in the prose: under odd vanishing, the
Euler characteristic (alternating sum) is a sum of non-negative terms
whenever the Betti numbers are non-negative. -/
theorem chi_nonneg (b :  $\mathbb{N} \rightarrow \mathbb{Z}$ ) (hodd :  $\forall j, b (2 * j + 1) = 0$ )
  (hpos :  $\forall j, 0 \leq b j$ ) (N :  $\mathbb{N}$ ) :
   $0 \leq \sum j \in \text{range } N, (-1 : \mathbb{Z}) ^ j * b j :=$  by
  rw [signed_eq_absolute b hodd N]
  exact Finset.sum_nonneg (fun j _ => hpos j)

end BettiParity

```

Certificate 7 — Appell–Lerch central coefficients (OddDivisors.lean)

```

/-
Certificate 7 (Proposition: Appell--Lerch central coefficients).
The central ( $y^0$ ) coefficient of the level-two Appell--Lerch sum
 $\mu(\tau, z) = \sum_{n \geq 1} q^n / ((1 - yq^n)(1 - y^{-1}q^n))$  at  $q^m$  counts the pairs
 $(n, a)$  with  $n \geq 1$ ,  $a \geq 0$  and  $n(2a+1) = m$  -- equivalently, the odd
divisors of  $m$ . This certificate machine-checks (i) the identity
 $\text{pairCount } m = d_{\text{odd}} m$  for ALL  $m \geq 1$ , via the explicit bijection
 $d = 2a+1$  with inverse  $(n, a) = (m/d, (d-1)/2)$ , and (ii) the
coefficient table to  $q^{15}$  by decidable evaluation. The
geometric-series expansion step from  $\mu$  to  $\text{pairCount}$  is a two-line
manipulation carried in the prose.
Unconditional; zero `sorry`.
-/
import Mathlib.NumberTheory.Divisors

namespace OddDivisors

open Finset

/-- The odd-divisor count  $d_{\text{odd}}(m)$ . -/
def dodd (m :  $\mathbb{N}$ ) :  $\mathbb{N} := ((\text{Nat.divisors } m).filter (fun d => Odd d)).card$ 

/-- The coefficient count: pairs  $(n, a)$ ,  $n \geq 1$ ,  $a \geq 0$ , with
 $n(2a+1) = m$ . Ranges suffice since  $n \leq m$  and  $(d-1)/2 \leq m$  for  $m \geq 1$ . -/
def pairCount (m :  $\mathbb{N}$ ) :  $\mathbb{N} :=$ 
   $((\text{range } (m + 1)) \times^s (\text{range } (m + 1))).filter$ 
   $(\text{fun } p => 1 \leq p.1 \wedge p.1 * (2 * p.2 + 1) = m)).card$ 

/-- **The identity, in general.** For all  $m \geq 1$ , the coefficient count
equals the odd-divisor count. -/
theorem pairCount_eq_dodd (m :  $\mathbb{N}$ ) (hm :  $1 \leq m$ ) :

```

```

    pairCount m = dodd m := by
  unfold pairCount dodd
  apply Finset.card_nbij' (fun p => 2 * p.2 + 1)
    (fun d => (m / d, (d - 1) / 2))
  · -- forward map lands in the odd divisors
    rintro (n, a) hp
    simp only [mem_filter, mem_product, mem_range] at hp
    obtain ((-, -), hn1, hprod) := hp
    simp only [mem_filter, Nat.mem_divisors]
    exact ((n, by rw [Nat.mul_comm]; exact hprod.symm), by omega),
      (a, rfl))
  · -- inverse map lands in the pair set
    intro d hd
    simp only [mem_filter, Nat.mem_divisors] at hd
    obtain ((hdvd, hm0), hodd) := hd
    have hd1 : 1 ≤ d := Nat.one_le_iff_ne_zero.mpr (by rintro rfl; simp at hodd)
    have hdle : d ≤ m := Nat.le_of_dvd (by omega) hdvd
    have hq1 : 1 ≤ m / d := Nat.one_le_div_iff (by omega) |>.mpr hdle
    have hqle : m / d ≤ m := Nat.div_le_self m d
    have hdodd : d % 2 = 1 := Nat.odd_iff.mp hodd
    simp only [mem_filter, mem_product, mem_range]
    refine ((by omega, by omega), hq1, ?_)
    rw [show 2 * ((d - 1) / 2) + 1 = d by omega]
    exact Nat.div_mul_cancel hdvd
  · -- left inverse
    rintro (n, a) hp
    simp only [mem_filter, mem_product, mem_range] at hp
    obtain ((-, -), hn1, hprod) := hp
    have hpos : 0 < 2 * a + 1 := by omega
    have hdiv : m / (2 * a + 1) = n := by
      rw [← hprod, Nat.mul_div_cancel _ hpos]
    simp only [Prod.mk.injEq]
    exact (hdiv, by omega)
  · -- right inverse
    intro d hd
    simp only [mem_filter, Nat.mem_divisors] at hd
    have hdodd : d % 2 = 1 := Nat.odd_iff.mp hd.2
    have hd1 : 1 ≤ d := by omega
    dsimp only
    omega

/-- **The table to q15**, by decidable evaluation. -/
theorem table :
  (List.range 15).map (fun i => dodd (i + 1)) =
  [1, 1, 2, 1, 2, 2, 2, 1, 3, 2, 2, 2, 2, 2, 4] := by decide

/-- The smallest case, restated: dodd(1) = 1. -/
theorem dodd_one : dodd 1 = 1 := by decide

/-- Strict positivity to q15, by decidable evaluation, accompanying
the one-line general proof in the prose. -/
theorem positive_to_fifteen : ∀ m ∈ List.range 15, 1 ≤ dodd (m + 1) := by
  decide

end OddDivisors

```

Certificate 8 — transit parity and the multiplier-weighted alternation (TransitParity.lean)

```

/-
=====
YM_A8_transit_parity.lean    (in-paper name: TransitParity.lean)
=====

```

Certificate 8 (transit parity and the multiplier-weighted alternation).

Part A (unconditional). On the eight-element quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$, modelled self-containedly as an explicit 4-component integer structure so that every finite claim is checked by `decide`: (i) the square of every pure imaginary unit is -1 , so the sign below is independent of the choice of axis; (ii) the action generated by the inner automorphisms by i and j together with the centre $\{-1\}$ decomposes Q_8 into exactly the four degree-2 orbits $\{\pm 1\}$, $\{\pm i\}$, $\{\pm j\}$, $\{\pm k\}$ – each pair closed under the generators, connected by the centre, pairwise disjoint, covering Q_8 , of cardinality two.

Part B (conditional). The two geometric identifications of the prose are encoded as named hypothesis fields: `h_lift` (each half-period translation of the torus lifts to the fibre as right-multiplication by a pure imaginary unit quaternion) and `h_wraps` (one full covering period equals two half-period translations, so the phase of the $(n+1)$ -th period multiplies the phase of the n -th by the square of the lift). Under exactly these hypotheses, the cumulative fibre phase over n full covering periods is $(-1)^n$.

Part C (unconditional). The single-term evaluation of the level-two multiplier-weighted generalised sum at modulus two, $K_2(n, -1; 2) = v^{-1} \cdot e^{\pi i(n-1)}$, with the multiplier value v^{-1} a parameter, satisfies $K_2(n, -1; 2) = (-1)^n$ for all $n \geq 1$ if and only if $v^{-1} = -1$; with trivial multiplier – the classical sum – the value is $-(-1)^n$ at every n , so the multiplier-weighted and classical objects genuinely differ.

Unconditional parts carry no hypotheses; zero `sorry`. No reference to any gauge group appears in this file.

Build: inside a Mathlib checkout (tag v4.15.0):
`lake env sh -c 'LEAN_PATH="$LEAN_PATH:$PWD" lean TransitParity.lean'`

```
=====
-/
import Mathlib.Data.Finset.Card
import Mathlib.Tactic.FinCases
import Mathlib.Tactic.Ring
import Mathlib.Analysis.SpecialFunctions.Trigonometric.Basic

noncomputable section
namespace TransitParity

/--! ## Part A – the quaternionic fibre algebra -/

/-- Integer quaternions, explicit 4-component model. -/
structure H4 where
  a : ℤ
  b : ℤ
  c : ℤ
  d : ℤ
deriving DecidableEq, Repr

namespace H4

/-- The quaternion product. -/
def mul (x y : H4) : H4 :=
  {x.a*y.a - x.b*y.b - x.c*y.c - x.d*y.d,
```

```

    x.a*y.b + x.b*y.a + x.c*y.d - x.d*y.c,
    x.a*y.c - x.b*y.d + x.c*y.a + x.d*y.b,
    x.a*y.d + x.b*y.c - x.c*y.b + x.d*y.a)

instance : Mul H4 := {mul}

def one : H4 := {1, 0, 0, 0}
def i : H4 := {0, 1, 0, 0}
def j : H4 := {0, 0, 1, 0}
def k : H4 := {0, 0, 0, 1}
def neg (x : H4) : H4 := {-x.a, -x.b, -x.c, -x.d}

/-- Embedding of the centre  $\mathbb{Z}$ . -/
def emb (m :  $\mathbb{Z}$ ) : H4 := {m, 0, 0, 0}

/-- The six pure imaginary units. -/
def pureUnits : Finset H4 := {i, neg i, j, neg j, k, neg k}

/-- The eight-point quaternion group  $Q_8$ . -/
def Q8 : Finset H4 := {one, neg one, i, neg i, j, neg j, k, neg k}

/-- Convention-independence: the square of EVERY pure imaginary
unit is -1, so the choice of axis for the half-period lift does
not affect the sign. -/
theorem sq_pure_unit_eq_neg_one :
   $\forall u \in \text{pureUnits}, u * u = \text{emb} (-1) := \text{by decide}$ 

/-- Generators of the action: conjugation by i, conjugation by j
( $i^{-1} = -i, j^{-1} = -j$  in  $Q_8$ ), and the centre. -/
def conjI (x : H4) : H4 := i * x * neg i
def conjJ (x : H4) : H4 := j * x * neg j
def negAct (x : H4) : H4 := neg x

/-- The four candidate orbits. -/
def orb1 : Finset H4 := {one, neg one}
def orbI : Finset H4 := {i, neg i}
def orbJ : Finset H4 := {j, neg j}
def orbK : Finset H4 := {k, neg k}

/-- Each candidate orbit is closed under all three generators
(hence under the generated group). -/
theorem orbits_closed :
  ( $\forall x \in \text{orb1}, \text{conjI } x \in \text{orb1} \wedge \text{conjJ } x \in \text{orb1} \wedge \text{negAct } x \in \text{orb1}$ )  $\wedge$ 
  ( $\forall x \in \text{orbI}, \text{conjI } x \in \text{orbI} \wedge \text{conjJ } x \in \text{orbI} \wedge \text{negAct } x \in \text{orbI}$ )  $\wedge$ 
  ( $\forall x \in \text{orbJ}, \text{conjI } x \in \text{orbJ} \wedge \text{conjJ } x \in \text{orbJ} \wedge \text{negAct } x \in \text{orbJ}$ )  $\wedge$ 
  ( $\forall x \in \text{orbK}, \text{conjI } x \in \text{orbK} \wedge \text{conjJ } x \in \text{orbK} \wedge \text{negAct } x \in \text{orbK}$ ) := by
  decide

/-- Within each orbit the two elements are connected by a
generator (the centre), so each closed pair is a single orbit,
not two. -/
theorem orbits_connected :
  ( $\forall x \in \text{orb1}, \text{negAct } x \neq x$ )  $\wedge$  ( $\forall x \in \text{orbI}, \text{negAct } x \neq x$ )  $\wedge$ 
  ( $\forall x \in \text{orbJ}, \text{negAct } x \neq x$ )  $\wedge$  ( $\forall x \in \text{orbK}, \text{negAct } x \neq x$ ) := by
  decide

/-- The four orbits partition  $Q_8$ : pairwise disjoint, covering,
each of cardinality exactly 2. -/
theorem orbits_partition :
  ( $\text{orb1} \cup \text{orbI} \cup \text{orbJ} \cup \text{orbK} = Q8$ )  $\wedge$ 
  ( $\text{Disjoint orb1 orbI} \wedge \text{Disjoint orb1 orbJ} \wedge \text{Disjoint orb1 orbK} \wedge$ 
     $\text{Disjoint orbI orbJ} \wedge \text{Disjoint orbI orbK} \wedge \text{Disjoint orbJ orbK}$ )  $\wedge$ 

```

```

    (orbI.card = 2 ∧ orbJ.card = 2 ∧ orbK.card = 2) := by
decide

/-! ### Arithmetic helpers -/

theorem emb_neg_one_mul (x : H4) : emb (-1) * x = neg x := by
  show mul (emb (-1)) x = neg x
  simp only [mul, emb, neg, H4.mk.injEq]
  refine {?, ?, ?, ?} <;> ring

theorem neg_emb (m : ℤ) : neg (emb m) = emb (-m) := by
  simp [neg, emb]

/-! ## Part B – the conditional transit parity -/

/-- The wrap data: the two geometric identifications of the prose,
as named hypothesis fields. -/
structure WrapData where
  /-- the fibre lift of one half-period translation -/
  halfLift : H4
  /-- identification (i): each half-period translation lifts as
  right-multiplication by a pure imaginary unit quaternion -/
  h_lift : halfLift ∈ pureUnits
  /-- the cumulative fibre phase after n full covering periods -/
  coveringPhase : ℕ → H4
  /-- identification (ii): one full covering period equals two
  half-period translations, so each period multiplies the
  cumulative phase by (halfLift · halfLift) -/
  h_wraps : ∀ n, coveringPhase (n + 1) =
    (halfLift * halfLift) * coveringPhase n
  h_zero : coveringPhase 0 = one

/-- **Certificate 8, Part B (the transit parity, conditional).**
Under the two identifications, the cumulative fibre phase over n
full covering periods is  $(-1)^n$ . -/
theorem transit_parity (D : WrapData) (n : ℕ) :
  D.coveringPhase n = emb ((-1) ^ n) := by
  induction n with
  | zero => rw [D.h_zero]; rfl
  | succ m ih =>
    rw [D.h_wraps, ih, sq_pure_unit_eq_neg_one D.halfLift D.h_lift,
      emb_neg_one_mul, neg_emb, pow_succ]
    congr 1
    ring

end H4

/-! ## Part C – the multiplier-weighted alternation -/

open Complex Real

/-- The level-two multiplier-weighted generalised sum at modulus
two: the units modulo 2 are {1}, so the sum has the single term
 $h = 1$  (with  $h \cdot \bar{h} \equiv -1 \pmod{2}$  giving  $\bar{h} = 1$ ), and the value is
 $v \cdot e^{2\pi i(n+1 + (-1) \cdot 1)/2} = v \cdot e^{\pi i(n-1)}$ , the multiplier
value  $v$  left as a parameter. -/
def K2c2 (nubar : ℂ) (n : ℕ) : ℂ :=
  nubar * Complex.exp (Real.pi * I * ((n : ℂ) - 1))

/-- The exponential factor evaluates to  $-(-1)^n$  for  $n \geq 1$ . -/
theorem exp_eval (n : ℕ) (h : 1 ≤ n) :
  Complex.exp (Real.pi * I * ((n : ℂ) - 1)) = -((-1) ^ n) := by

```

```

obtain ⟨m, rfl⟩ := Nat.exists_eq_add_of_le h
have h1 : ((1 + m : ℕ) : ℂ) - 1 = (m : ℂ) := by push_cast; ring
rw [h1, show Real.pi * I * (m : ℂ) = (m : ℂ) * (Real.pi * I) by ring,
    Complex.exp_nat_mul, Complex.exp_pi_mul_I]
rw [pow_add, pow_one]
ring

/-- **Certificate 8, Part C (the characterisation).** The
alternation  $K_2(n, -1; 2) = (-1)^n$  holds for all  $n \geq 1$  if and only
if the multiplier value  $v$  equals  $-1$ . -/
theorem characterisation (nubar : ℂ) :
  (∀ n : ℕ, 1 ≤ n → K2c2 nubar n = (-1) ^ n) ↔ nubar = -1 := by
  constructor
  · intro h
    have h1 := h 1 le_rfl
    simp only [K2c2] at h1
    rw [exp_eval 1 le_rfl] at h1
    simp using h1
  · intro h n hn
    simp only [K2c2, h, exp_eval n hn]
    ring

/-- With trivial multiplier – the classical sum – the value is
 $-(-1)^n$  at every  $n \geq 1$ : the multiplier-weighted and classical
objects genuinely differ, by an overall sign at every  $n$ . -/
theorem classical_sum_differs (n : ℕ) (hn : 1 ≤ n) :
  K2c2 1 n = -((-1) ^ n) := by
  simp only [K2c2, one_mul]
  exact exp_eval n hn

end TransitParity

```

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